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## Research article

## Loggers affect the foraging behaviour and fitness of European shags

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Biologgers enable real-time collection of detailed behavioural and physiological data from wide-ranging animals, including seabirds inhabiting remote regions. However, the number of studies using tracking devices has not been matched by research exploring the behavioural and fitness costs of deployment, and the effects on data reliability. We assessed how GPS loggers, video loggers, and time depth recorders (TDRs) affect the behaviour, physiology, and reproductive performance of European shags *Gulosus aristotelis* breeding on Sklinna, Norway. The loggers varied in mass, attachment location and deployment duration, allowing comparison of their relative effects. Birds without loggers served as controls to assess logger-related changes in adult body mass, chick growth, reproductive success, and survival. Birds with longer-term tail-mounted GPS loggers and leg-mounted TDRs showed altered foraging behaviour, including shorter trips and dives, compared to birds with tail-mounted GPS and TDRs of the same weight, instrumented for only two days. A mean loss in body mass was experienced by adult birds regardless of the logger type used, while chick growth rate dropped to 20% of that observed in control nests when video, TDR and GPS loggers (~ 4% of body mass) were deployed together. Logger attachments did not impact reproductive success, and overall logger birds showed higher survival than controls. However, female survival was lower than that of males among birds fitted with video and long-term GPS loggers. Our results demonstrate the importance of measuring behavioural and physiological effects that can scale over time. The advances in our understanding of animal ecology and behaviour generated by biologging have been impressive, but there is a need to consider the impacts on animal welfare and data quality. Consistent reporting of logger deployment details is essential to assess biologging impacts across species and refine protocols that account for device weight, drag and attachment location.

Keywords: biologging, conservation, monitoring, movement ecology, seabird, tag effects



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## Introduction

Movement is central to animal ecology and evolution, influencing species that move regularly throughout their lives as well as species that only move during specific life history stages (Nathan et al. 2022). It also provides a key metric for exploring how animals respond to rapid environmental changes, including those increasingly driven by human activities (Doherty et al. 2021). The field of movement ecology has grown rapidly in recent decades as technological advances have led to the development and miniaturisation of biologging sensors (herein 'loggers') that collect detailed data on animal location, physiological state, social interactions and the environmental conditions to which an animal is exposed (Kays et al. 2015, Wilmers et al. 2015, Watanabe and Papastamatiou 2023). Indeed, the field has moved from one where technological constraints and limited data were major restrictions to the challenge of storing, processing and analysing large datasets collected at increasingly fine-scale resolutions (Nathan et al. 2022).

Detailed animal movement data have revealed fascinating ecological and behavioural insights, particularly for elusive and wide-ranging species (Chung et al. 2021). This is typified by marine animals, such as seabirds, that forage over large areas. They often spend a considerable amount of time at sea and are therefore difficult to observe when not at their colony (Yoda 2019). Findings from movement research are being used to inform marine spatial planning by identifying key foraging habitats and biodiversity hotspots (Soanes et al. 2016, Péron et al. 2018), as well as to assess the impacts of human activities such as fishing (Cleasby et al. 2022) and the rapid development of offshore infrastructure, e.g. wind farms (Thaxter et al. 2019). Meanwhile, global datasets on the movements of petrels and albatross have been used to determine the importance of national jurisdictions and the connectivity between them, facilitating collaborative conservation efforts for these transoceanic species (Beal et al. 2021).

These discoveries and the crucial importance of movement to seabird ecology and conservation have seen a surge in the number of loggers deployed. Moreover, miniaturisation of the technology has enabled an increasing range of species to be studied (Portugal and White 2018), while developments in power supply (e.g. greater battery life and incorporation of solar panels) have enabled tracking over longer periods (Gauld et al. 2023). However, the deployment of loggers can potentially alter individual behaviour and impact fitness, including compromised survival and reduced breeding success (Barron et al. 2010, Bodey et al. 2018, Geen et al. 2019). These effects are of welfare and conservation concern, particularly for species that already face considerable anthropogenic pressures. Additionally, there are implications for study rigour and inference if the behaviour being observed is influenced by the method being used to collect the data.

Over the past two decades, an evidence-based guideline has been used in seabird research suggesting that loggers should be a maximum of 3% of the bird's mass. This was

derived from a review of twenty studies exploring the effects of loggers on albatrosses and petrels (Phillips et al. 2003). However, in addition to the cost of the weight of the logger, birds that rely on diving and swimming to catch prey may also suffer from increased hydrodynamic and aerodynamic drag (Vandenabeele et al. 2012, 2015, Clairbaux et al. 2025). Nevertheless, with rapid technological advancements and miniaturisation of these devices, the mass-based rules continue to be used to justify the deployment of loggers across an increasing range of species with varied morphology, ecology and behaviour (Portugal and White 2018).

A phylogenetically controlled meta-analysis across all groups of birds (land and sea) revealed that, although small, the negative effects of loggers on survival, reproduction and parental care were detectable when devices exceeded 1% of body mass (Bodey et al. 2018). There was also a greater effect of loggers for birds that used flapping flight versus those that more commonly soared, e.g. procellariiforms. Despite these findings, a considerable number of studies still fail to report sufficient information to assess the impacts of logger deployment. A review by Geen et al. (2019) reported that less than 50% of tracking studies detailed the effects of device deployment. Furthermore, there are comparatively few studies that have explicitly explored the effects of loggers on bird flight and fitness, particularly when compared with the rapid increase in the number of publications arising from these studies (Geen et al. 2019). A key challenge in assessing the effects of logger deployment is establishing a true control, especially when studying behaviour at sea. Researchers have addressed this by comparing different loggers (Gillies et al. 2020), varying deployment duration, and examining demographic factors like age and sex to assess potential impacts (Barron et al. 2010, Bodey et al. 2018).

In this study, we explored the effects of logger deployments on the behaviour and reproductive performance of European shags *Gulosus aristotelis* (hereafter 'shag'), a large, nearshore-foraging seabird that exhibits sexual dimorphism (Velando et al. 2000), with males at our study site being approximately 15% heavier than females (male:  $1942 \pm 123$  g; female:  $1691 \pm 113$  g; mean  $\pm$  SD). Shags are foot-propelled divers that keep their wings tucked by their side to reduce drag when underwater. Their webbed feet and powerful leg strokes enable them to dive vertically from 5 to 60 m in depth (Watanuki et al. 2005, Christensen-Dalsgaard et al. 2017). The foot-propelled diving style of shags makes them particularly relevant when considering the effects of leg-mounted loggers. We used an extensive, multi-year tracking dataset of shags breeding at Sklinna, central Norway (Fig. 1), in combination with data on breeding success, chick growth and post-deployment survival of tracked and untracked adult birds. Tracking was conducted using two types of GPS loggers, one type of temperature-depth recorder (TDR), and two types of video loggers (Table 1). Birds were instrumented typically for either > 10 days (tail-mounted GPS, leg-mounted TDR), or for 1–3 days (tail-mounted GPS and TDR, with or without a back-mounted video-logger). In shags, both parents provision chicks (Cramp and Simmons 1977), and approximately

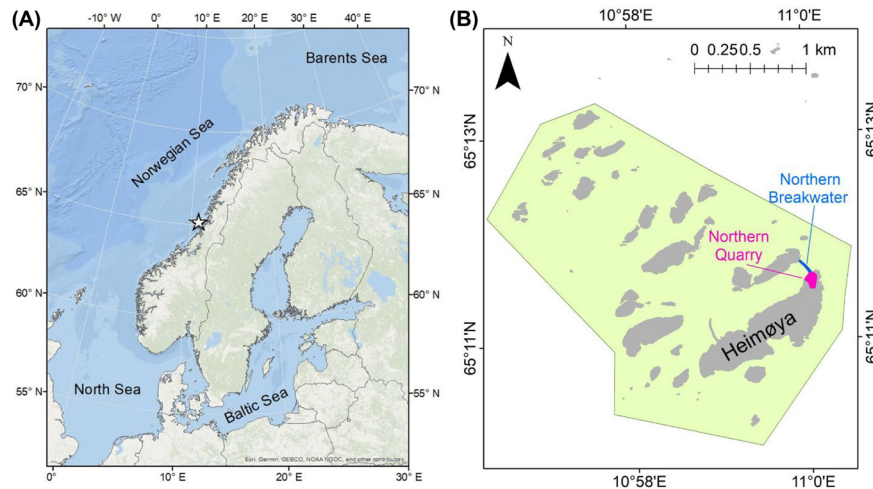


Figure 1. (A) The location of Sklinna in central Norway, and (B) the study areas on the main island Heimøya. Land is shown in grey, while the protected area is marked as a green polygon.

equal numbers of males and females were equipped with each category of logger.

The study aimed to explore whether logger attachments affected foraging trip metrics, dive behaviour, chick growth and reproductive performance of adults (breeding success, adult body mass and adult survival post deployment). We predicted that: 1) the behavioural and reproductive performance impacts of logger attachment would increase as a function of logger mass, drag, and deployment duration, driven by greater energetic costs; 2) dive behaviour would be affected to a greater extent in birds with leg-mounted TDRs compared to tail-mounted, as shags are foot-propelled divers; 3) chick growth would be compromised to the greatest extent by the deployment of the largest loggers affecting adult provisioning; 4) females, being smaller than males but fitted with the same loggers, would experience greater impacts; and 5) these effects would be more pronounced in years with lower food availability.

## Material and methods

### Study site

Field work was conducted during the annual breeding season (June–July) from 2011–2024 on Heimøya, an island in the Sklinna archipelago (65°12'N, 10°59'E) (Fig. 1A). The colony is part of the SEAPOP monitoring programme ([www.seapop.no/en](http://www.seapop.no/en)). During the study period, between 1257 and 2570 (average  $1906 \pm 363$  SD) shags were breeding on Heimøya. Work was carried out in two adjacent sections of the island: The inner part of the northern breakwater (124–283 nests; average:  $205 \pm 54$  SD), and the adjacent northern quarry (61–202 nests; average  $110 \pm 40$  SD) (Fig. 1B). All GPS tracking was performed on the inner part of the breakwater, while we monitored approximately 50 control nests in the quarry area, where we assessed reproductive success and chick growth.

### Logger deployments

Loggers were deployed on adult birds from a randomly selected subset of nests where individuals tolerated researcher approach, either during incubation (PathTrack loggers only) or chick-rearing, when chicks were between 1 and about 25 days old (all other logger types). We captured adults by hand or with a noose pole. Sex was determined by size and vocalizations (Cramp and Simmons 1977). If birds did not already have a metal ring, they were ringed during deployment. Five types of loggers were used in this study: IgotU GT-120 GPS loggers (Mobile Action Technology, Taiwan), G5 time depth recorders (TDRs), nanoFix® GEO+RF GPS loggers (PathTrack, UK), DVL400M065-VD3GT video loggers (Little Leonardo, Italy), and GipsyCAM video loggers (TechnoSmArt, Italy), along with dummy video loggers for testing attachment methods (Table 1). Loggers were attached to birds upper back, tail and/or leg, depending on the model used (see Table 1 and Fig. 2 for attachment details). Birds without loggers, fitted only with a metal ring, were used as a control group for the body mass, chick growth and survival analyses.

Loggers were deployed in three main regimes (hereafter: logger categories) between 2020 and 2022: IgotU deployments (IgotU GPS+TDR), PathTrack deployments (PathTrack GPS+TDR), and video deployments (video loggers+IgotU GPS+TDR). The total weight of loggers, including attachment materials, the percentage of body mass they represented as well as attachment location (tail-, back- and/or leg-attachment) are summarised in Table 1. 'Tail-mounted' refers to devices affixed to the three or four central tail feathers using waterproof Tesa tape, a standard method for attaching loggers to seabirds (see Fig. 2). Tape was used consistently across all logger deployments except leg-mounted TDRs, which were mounted with cable-ties and superglue to a plastic ring. Adult birds were recaptured within 1–3 days to recover video and IgotU loggers, and within 3–8 days ( $3.7 \pm 1.5$  SD) for short-term PathTrack



Table 1. Details of the logger categories, number of birds, logger sizes and deployments at Heimøya, Sklinna (2020–2022). For analyses of foraging, diving, chick growth, breeding success, and adult body mass, we classified the data using the three logger categories (first column). For adult survival analysis, deployments were categorised as short- or long-term based on duration and logger mass (first column).  $n$  = total sample size;  $M$  = no. of males;  $F$  = no. of females.

| Logger category (behavioural/survival impacts) | Logger combination                                                 | Sample size                 | Logger dimensions                                                                                                                                                                                                     | Total logger weight (incl. tape) | Body mass (%), mean $\pm$ SD               | Deployment duration                                                                                               | Deployment periods                                                               |
|------------------------------------------------|--------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| IgotU/short-term deployment                    | IgotU GPS+TDR (both taped to tail)                                 | $n=180$<br>$M=87$<br>$F=93$ | IgotU GT-120 (50 $\times$ 26 $\times$ 12 mm, 23 g) + G5 TDR (35 $\times$ 12 mm, 6.5 g)                                                                                                                                | 32.7 $\pm$ 1.0 g                 | $M=1.70 \pm 0.10\%$<br>$F=2.00 \pm 0.13\%$ | 48–72 hours                                                                                                       | 19 June–11 July 2020; 23 June–07 July 2021; and 24 June–07 July 2022             |
| PathTrack/long-term deployment                 | PathTrack GPS (taped to tail) + TDR (attached to leg-ring)         | $n=30$<br>$M=15$<br>$F=15$  | PathTrack nanoFix GEO+RF (60 $\times$ 25 $\times$ 25 mm, 21.8 g) + G5 TDR                                                                                                                                             | 33.4 $\pm$ 1.0 g                 | $M=1.80 \pm 0.11\%$<br>$F=2.00 \pm 0.13\%$ | GPS: 10–87 days (avg. 59 $\pm$ 22 SD), TDR removed by mid-July (GPS+TDR combined: 10–31 days, avg. 23 $\pm$ 8 SD) | 24 June–22 September 2020; 11 June–25 August 2021; and 11 June–06 September 2022 |
| PathTrack/short-term deployment                | PathTrack GPS (taped to tail) + TDR (attached to leg-ring)         | $n=10$<br>$M=5$<br>$F=5$    | PathTrack nanoFix GEO+RF (60 $\times$ 25 $\times$ 25 mm, 21.8 g) + G5 TDR                                                                                                                                             | 33.4 $\pm$ 1.0 g                 | $M=1.80 \pm 0.15\%$<br>$F=2.00 \pm 0.09\%$ | GPS and TDR deployed for 3–8 days (avg. 3.7 $\pm$ 1.5 SD)                                                         | 19–27 June 2020                                                                  |
| Video/short-term deployment                    | Video logger (taped to upper-back) + IgotU GPS+TDR (taped to tail) | $n=27$<br>$M=14$<br>$F=13$  | DVL400M065-VD3GT (70 $\times$ 20 $\times$ 14 mm, 30.5 g)/Dummy DVL400M065 (70 $\times$ 20 $\times$ 14 mm, 24.5 g)/GipsyCAM (71 $\times$ 24 $\times$ 5 to 71 $\times$ 24 $\times$ 25 mm, 35 g) + IgotU GT-120 + G5 TDR | 67.5 $\pm$ 1.0 g                 | $M=3.50 \pm 0.28\%$<br>$F=4.10 \pm 0.21\%$ | 24–48 hours                                                                                                       | 24 June–9 July 2020; and 23 June–04 July 2021                                    |

deployments. For long-term PathTrack deployments, birds were recaptured after 10–31 days ( $23 \pm 8$  SD) to retrieve the TDR, while the GPS logger remained in place for 10–87 days ( $59 \pm 22$  SD). In cases where technical issues occurred with the GPS, it was retrieved along with the TDR before the scheduled end period (Table 1).

Handling time during deployment depended on logger category (3–12 min), with IgotU deployments typically being fastest to deploy, taking under 5 minutes. Birds returned to their nests within a few minutes of release. Due to limited nest accessibility, we frequently deployed video and IgotU loggers (but not PathTrack loggers) on both partners of a nest. Deployments were either consecutive, where one bird's logger was removed before the partner was tagged, or partly overlapping, with both partners carrying loggers simultaneously for 24–48 h. In approximately half of all nests, both partners were tagged, with the order of male and female deployment assigned at random. This applied to IgotU (59 of 118 deployments, 2020–2022) and video loggers (18 of 32 deployments, 2020–2021), but not PathTrack loggers (only one partner per nest per season).

IgotU loggers recorded positions at 1-minute intervals, TDR loggers at 1-second intervals, video loggers recorded continually for up to 5 hours, and PathTrack loggers recorded data at intervals of either  $\geq 30$  seconds (in a short 2020 trial) or  $\geq 5$  minutes (for most deployments). PathTrack data were downloaded via UHF to a base station located within 500 m of the nests. As the battery depleted, the GPS sampling rate automatically decreased in steps (e.g. from 5 to 10, 15, 20 minutes, and so on).

### Adult body mass and chick measurements of logger birds

Because shags have negligible fat reserves, we used within-individual changes in body mass as a measure of body condition (De Vries and Van Eerden 1995, Enstipp et al. 2007). We measured the body mass of adult shags to the nearest 10 g using a Pesola spring scale during logger deployments. While adults were handled and deployed with loggers, the chicks were removed from the nest (if reachable), and the number of chicks and eggs as well as the body mass of chicks was recorded. Chicks were weighed with Pesola spring scales and accuracy of mass was dependent on chick size (to the nearest g if chicks were  $< 100$  g; to the nearest 5 g if chicks were 100–1000 g; to the nearest 10 g if chicks were  $> 1000$  g). During the second capture, the adult and chick body masses were measured again. Adults received an individually coded colour ring during the second capture if they had not previously been equipped. Variation in stomach contents could influence adult and chick body mass measurements. However, given the large sample size and multi-year dataset, we don't expect this biased overall patterns.

### Reproductive success, chick growth and adult body mass of control nests

Over the 14-year period (2011–2024), we annually monitored 46–54 control nests of birds that were not fitted with

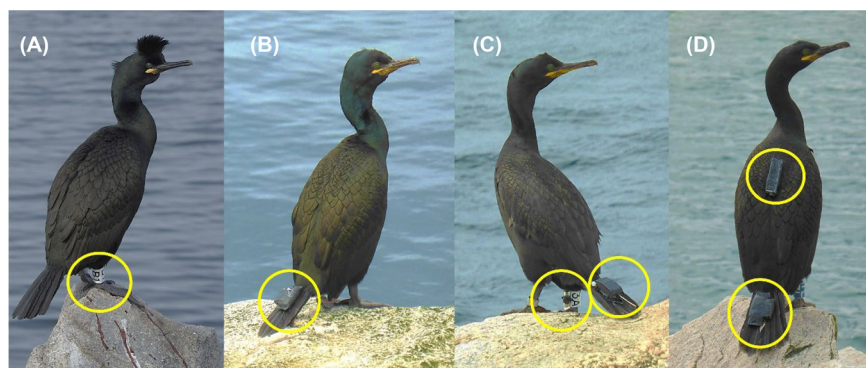


Figure 2. Logger categories and attachment locations on European shags *Gulosus aristotelis*: (A) ring only; (B) IgotU GPS with TDR, both attached to the tail; (C) PathTrack GPS attached to the tail with TDR mounted on the leg ring; (D) video logger attached to the back, with GPS and TDR on the tail. All photos: Nina Dehnhard

loggers. These were checked every fifth day between 5 June and 15 July. Visit intervals were occasionally four or six days due to adverse weather. Accessible nests were selected regardless of parental approachability and marked with numbered tape, which were removed after the final visit. Body mass was recorded as described for chicks of GPS-logger birds.

Shags exhibit hatching asynchrony, with the first laid egg hatching first and the chick growing fastest (Stokland and Amundsen 1988). However, egg laying order was usually unknown as it happened before our arrival in the colony. At each visit, we therefore assigned the largest and heaviest chick in the nest as number 1, the next largest number 2 and so on. Given the hatching asynchrony and size differences among the shag chicks in the same nest, we deemed this approach sufficient until we could mark chicks with numeric metal rings at an age of about 17 days).

### Adult capture-mark-resight data

We used individual-level capture-mark-resight (CMR) data collected from 2015 to 2024 at the northern breakwater ( $n_{\text{individuals}} = 387$ ), including both logger and non-logger individuals (Supporting information) to estimate survival probabilities. All control birds (without a logger in a given year) were colour-ringed and had been deployed with a logger in at least one previous year (possibly before 2015). IgotU loggers were used 2011–2023, video loggers in 2020–2021 and long-term PathTrack loggers in 2020–2023. Each year during the field season (6–7 weeks annually: one week in early May and six weeks from early June to mid-July), the northern breakwater was visited almost daily to resight colour-ringed individuals using binoculars, a super-zoom camera, or a telescope. Also, any adjacent areas and rocks where shags roosted close to the breakwater were screened for individuals with coded colour rings.

### Data analysis

All data analyses were conducted in R ver. 4.4.2 ([www.r-project.org](http://www.r-project.org)). The first stage of the analysis focussed on foraging trip length and duration, and dive behaviour. The second stage of the analysis focussed on the demographic and physiological metrics, which are directly linked to reproductive

performance, including change in adult body mass, chick growth rate, reproductive success and adult survival.

### Treatment of GPS and dive data

We used GPS data to measure the distance of foraging trips from the colony, and dive data to investigate fine-scale foraging behaviour. These data were treated in a similar way to previous research conducted at our study site (Christensen-Dalsgaard et al. 2017, Lorentsen et al. 2019, Dehnhard et al. 2022). We excluded locations within 500 m of the colony, since these locations are mostly associated with resting and preening activities as well as washing dives, but not with foraging activities. Apparent locational outliers were removed using a speed filter with a maximum speed of  $30 \text{ m s}^{-1}$ , and a speed filter of  $15 \text{ m s}^{-1}$  on strongly curvaceous flight paths, as described in Lorentsen et al. (2019). Following Lorentsen et al. (2019), we defined foraging trips as movement paths  $\geq 5 \text{ min}$   $\geq 500 \text{ m}$  away from the colony. For comparing foraging trip metrics, we excluded incomplete foraging trips: i.e. trips where: 1) locations at the colony were not available either before or after the trip, 2) gaps of  $> 30$  minutes (to account for the GPS-intervals of PathTrack loggers) existed between the first (or last) location in the trip and the previous (or next) location at the colony, and 3) gaps of  $> 60$  minutes existed between locations during the trip. Due to the inability of GPS loggers to acquire locations when submerged (i.e. when the bird is diving), trips recorded by both GPS logger types frequently included gaps between locations. The one-hour cut-off to define incomplete trips was chosen as a conservative measure based on average trip duration and the obtained GPS-intervals.

TDR-logger files were calibrated and summarised using the 'diveMove' package (Luque 2007). Dives shallower than 2 m were excluded as likely washing dives (Christensen-Dalsgaard et al. 2017, Dehnhard et al. 2022). Surface calibration employed the zoc method (Luque 2007) with three sequential filters – a 3 s median smoothing filter, a 3 s 0.1 quantile filter, and a 60 s 0.02 quantile filter – constrained to depths between  $-4$  and  $4 \text{ m}$ . For each dive, average and maximum depth, dive duration, and bottom time (i.e. total dive time – (descent time + ascent time)) were computed.

Dives with speeds exceeding  $3 \text{ m s}^{-1}$  were deemed unlikely (Watanuki et al. 2008) and removed from the dataset.

### Foraging trip and dive behaviour

Foraging behaviour data were collected from loggers during the chick-rearing season from 2020 to 2022. Since PathTrack deployments were longer and started earlier than the other logger types, datasets were standardised by focusing on a two-week period (23 June–7 July) at the peak of chick hatching. This timeframe matched the deployment period of the other loggers, allowing for direct comparison of foraging behaviour. Although this reduced data availability, it ensured consistent environmental conditions and breeding behaviours across years and logger types, while balancing sample sizes of trip and dive parameters between IgotU and PathTrack loggers.

A generalised linear mixed modelling approach was used to analyse five different metrics of the foraging behaviour: maximum distance from the colony, foraging trip duration, maximum dive depth, total dive time and bottom time. Data exploration indicated that a logarithmic transformation was required for trip duration, maximum dive depth, total dive time and bottom time. The identity of the tagged bird was included as a random effect, as multiple foraging trips were recorded for each individual. Four candidate models were run for each of the five response variables using logger category (short-term and long-term PathTrack deployments were pooled for these analyses), sex, year and time since logger deployment (in days) as explanatory variables. Model one was an additive model including all four variables, while model two excluded the time since deployment. Model three included all of the variables and an interaction term (logger category  $\times$  year) to assess if foraging behaviour differed as a function of logger type across the three years, particularly as there was a decline in saithe *Pollachius virens* abundance (a key prey species) during 2021 and 2022 (ICES 2024), which can affect both foraging efficiency and reproductive success in shags (Lorentsen et al. 2019). Model four contained all of the variables and two interactions (logger category  $\times$  year and logger category  $\times$  sex) to explore both year and sex differences on the relative effects of the loggers (Supporting information). The AIC score and AIC weight was used to rank the models and determine which model best explained the data.

We validated our models by assessing residuals for normality and homoscedasticity using residual plots, histograms, and Q–Q plots with reference lines. Multicollinearity among predictors was evaluated using variance inflation factors (VIF), with high values indicating potential redundancy. To identify influential data points, we calculated Cook's distance and compared values to identify points that might disproportionately affect model estimates.

### Adult body mass and chick growth rates

Logger birds were weighed before and after deployment. The body mass measurements taken at logger deployment were used as control data, representing each bird's condition prior to carrying a device. Therefore, these 'control' measurements consisted of a single pre-deployment value per individual,

collected from early June to mid-July across multiple years. In order to assess body mass change over time, each mass was paired with its deployment date and converted to day-of-year (DOY). We modelled the relationship between body mass and DOY using sex-specific linear regressions to characterise natural seasonal variation in body mass during the breeding period. Predicted body masses at the end of the deployment window (15 July) were extracted from the model to estimate the expected seasonal mass change in the absence of logger effects. We used a generalised linear model (GLM) to compare observed mass changes across logger types (IgotU, PathTrack, Video) and the predicted values for the control group, accounting for sex as a covariate.

The daily growth rate of chicks was calculated across the control nests using the body mass data that was collected every five days over a 25-day pre-fledging period (2011–2024). These data were also used to confirm that body mass increased linearly from day 5–30 and that there was no significant difference in growth rate between years (Supporting information). As such, we could be confident that the two measurements taken for the chicks in nests of logged birds (pre and post deployment) could be reliably used to determine a daily growth rate. The data were filtered to remove individuals that showed a change in body mass that exceeded  $100 \text{ g day}^{-1}$  (presuming measurement errors or typos in the field). A GLM was used to explore whether there was a difference in growth rate between the chicks of control nests compared with those at the treatment nests – where an adult was deployed with one of the three logger categories (IgotU: 2012–2023, PathTrack: 2020–2022, video: 2020–2021).

The data for the models described above were explored using summary statistics and visualisations to assess the distribution of the response variable, detect outliers, and handle missing values. Potential collinearity between predictors was explored using contingency tables. For model validation, residual diagnostics were performed, including plots of residuals versus fitted values and Q–Q plots, to assess assumptions such as normality and homoscedasticity, while standardised residuals and Cook's distance were examined to identify potential outliers and influential observations.

### Reproductive success

The proportion of chicks that survived the annual survey period (2020–2022) was calculated for each of the control nests and for nests where an adult was fitted with a long-term PathTrack logger. The other logger categories were omitted from this analysis due to short deployment durations. Nest content of PathTrack birds was registered, and chicks measured (whenever possible) at the first deployment, and subsequently when TDRs were removed towards mid-July, on average 23 days ( $\pm 8 \text{ SD}$ ) later. For control nests ( $n = 139$ ), the survey period was longer (spanning 40 days; 5 June–14 July). We cropped the survey period for control nests to the closest visit dates (from  $\sim 15$  June to  $\sim 10$  July) to match it with the dataset obtained from PathTrack birds, thus limiting the data to 24 days ( $\pm 2 \text{ SD}$ ). To account for the different clutch sizes (1–4 eggs) and the bounded nature of proportion



data (constrained between 0 and 1), we calculated an adjusted proportion of chicks surviving per nest using the following equation where  $p$  is the observed proportion of chicks that survived, and  $n$  is the total clutch size:

$$\text{adjusted proportion} = \left( p \times (n - 1) + 0.5 \right) / n$$

Beta regression models with a logit function were used to analyse reproductive success as they are well-suited to handling proportion data. Model coefficients are reported on the logit scale, while we also provide mean survival probabilities from the raw data. A main-effects model was fitted with logger status (control vs treatment) and year as predictors, using 2022 as the reference category. Quantile residuals were used to confirm model fit.

### Adult survival estimation

To estimate the effects of loggers on adult survival from one breeding season to the next, loggers were categorised based on deployment duration and weight into three groups: 1) short-term loggers (IgotU and PathTrack, deployed for less than 8 days), 2) video loggers (deployed for 1–3 days), and 3) long-term loggers (PathTrack, deployed for more than 28 days). These categories were chosen as deployment duration and logger mass were expected to have the greatest influence on survival probability, while TDR placement, total logger mass, and drag were likely to be more critical for foraging behaviour. From 2015–2024, there were 879 annual observations of individually colour-ringed birds without a logger, 300 of colour-ringed individuals with a short-term logger, 28 of colour-ringed individuals with video loggers and 32 of colour-ringed individuals with long-term loggers (Supporting information).

Analysis of the CMR data was based on a Cormack–Jolly–Seber (CJS) model framework (Lebreton et al. 1992). The goodness of fit (GOF) of this model to the data was assessed using UCARE (Choquet et al. 2009). One of the GOF test components indicated a significant deviation from the model assumptions: Test 2.CT showed trap-dependence (dependence of resighting rates across years,  $\chi^2 = 26.534$ ,  $df = 5$ ,  $p < 0.001$ ). We corrected for this by including a trap-dependence effect in the resighting model ( $p$ ), where resighting rates of individuals the year after first capture are estimated separately from those that were not resighted the previous year. We formulated a matrix (Trap), which indicates whether individual  $i$  was caught the year prior (year  $t$ ) or not.

A separate intercept was estimated for each level of ‘logger’, which was fitted as individual- and time-varying covariate and indicates whether individual  $i$  at year  $t$  had either 1) no logger, 2) a short-term logger, 3) video logger or 4) long-term logger attached. Although alternative models were available, e.g. multi-state models, we opted for estimating logger effects via individual- and time-varying covariate to avoid the estimation of multiple transition probabilities, which were not considered biologically relevant to this study. Given the sexual dimorphism of European shags, sex was included

as an interactive effect with logger to test whether logger effects were greater for females given their smaller body size. However, we did not expect a baseline difference in survival between sexes and so only the interactive effect was included. The survival and observation models were specified as:

$$\text{Logit}(\phi_{i,t}) \sim \alpha_{\phi} [\text{Logger}_{i,t}] + \beta_s [\text{Logger}_{i,t}] \text{Sex}_i$$

$$\text{Logit}(p_{i,t}) \sim \alpha_p [\text{Trap}_{i,t}] + \varepsilon_{p,t}$$

A separate model was fitted to estimate annual survival rates, without the sex-logger interaction, as a model with both a sex-logger interaction and year random effect did not sufficiently converge ( $\hat{R}$  values for  $\varepsilon_{s,t}$  were greater than 1.1). The recapture model was the same as above.

$$\text{Logit}(\phi_{i,t}) \sim \alpha_{\phi} [\text{Logger}_{i,t}] + \varepsilon_{s,t}$$

The model was fitted in a Bayesian framework using MCMC simulations to estimate posterior distributions of model parameters. Following Gelman et al. (2008) we used weakly informative default priors for parameters to aid convergence. Normal distributions were specified for intercepts and regression coefficients and gamma distributions for the variance ( $\sigma$ ) terms (prior distributions are summarised in the Supporting information). Three chains were run of length 200 000, where the first 50 000 was discarded as burn-in. The model was implemented in JAGS (Plummer et al. 2003), via the program JAGSUI, ver. 1.5.2 (Kellner 2015) in R ver. 4.2.2 ([www.r-project.org](http://www.r-project.org)). Convergence was assessed by ensuring each parameter's  $\hat{R}$  value was less than 1.1 (Brooks and Gelman 1998).

## Results

### Foraging trip metrics

Model selection supported the inclusion of logger, sex, time since logger deployment, and a logger-year interaction for the analysis of maximum distance from the colony, with the top model (model 3) contributing all the AICc weight (Supporting information). Birds fitted with PathTrack loggers had significantly shorter trips in 2021 ( $\beta = -18475.64$ ,  $p < 0.001$ ) and 2022 ( $\beta = -17725.25$ ,  $p < 0.001$ ) compared with 2020 (Fig. 3A). However, overall foraging trips were significantly longer in distance in 2021 ( $\beta = 9105.76$ ,  $p < 0.001$ ) and 2022 ( $\beta = 6908.60$ ,  $p < 0.001$ ) compared to 2020. Increasing time since logger deployment was also associated with greater maximum distance ( $\beta = 728.35$ ,  $p < 0.001$ ).

Model 2 (logger, year and sex) was the best-fitting model for the analysis of trip duration and accounted for 98% of the AICc weight (Supporting information). The model revealed that compared to birds with IgotU loggers, those with video

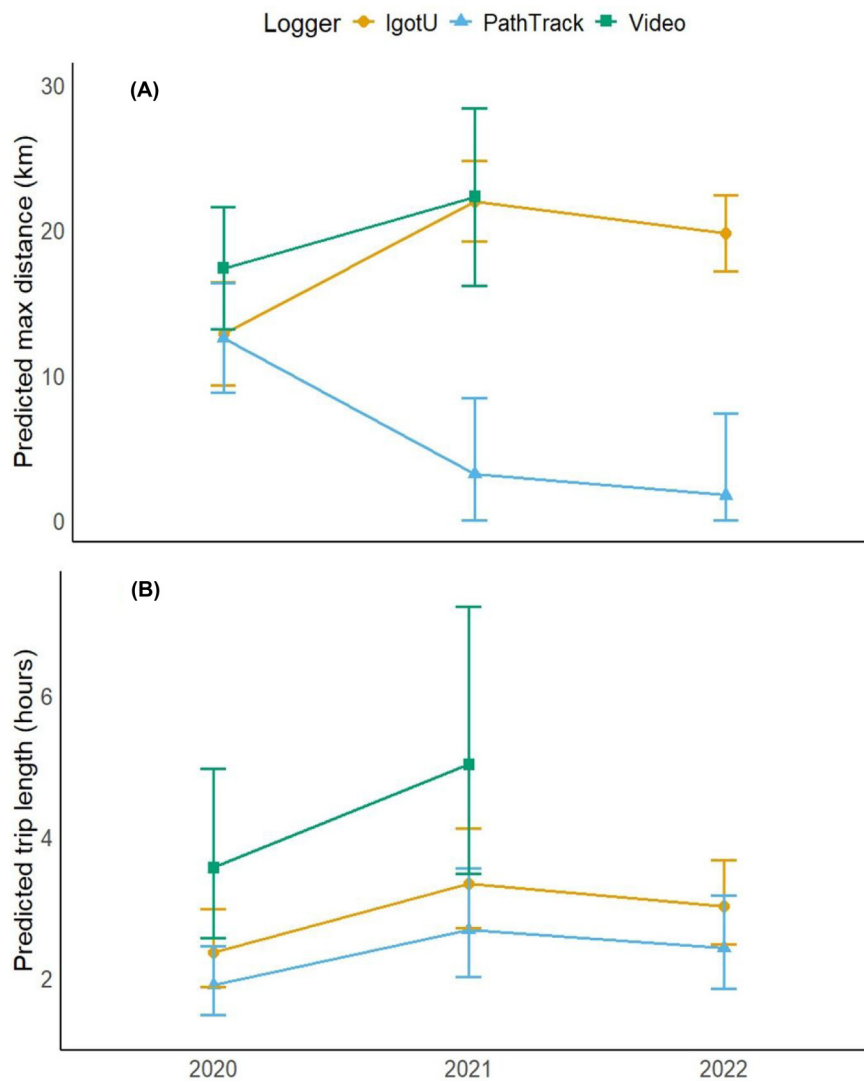


Figure 3. Predicted values from generalised linear mixed models: (A) maximum distance of foraging trips from the colony by logger type and study year, and (B) duration of foraging trips by logger type and study year. Error bars show 95% confidence intervals; negative lower bounds are truncated at zero, as negative distances are not biologically meaningful.

loggers undertook longer foraging trips ( $\beta = 0.411$ ,  $p = 0.024$ ) while those with PathTrack loggers showed a trend toward shorter trips ( $\beta = -0.219$ ,  $p = 0.084$ ; Fig. 3B). Trip durations were also significantly longer in 2021 ( $\beta = 0.343$ ,  $p = 0.006$ ) compared with 2020, while there was also evidence of an increase in 2022 ( $\beta = 0.242$ ,  $p = 0.061$ ). Since the interaction term with sex and logger type was not retained in the top models for either of the foraging trip metrics, there was no evidence that the loggers affected the sexes differently.

### Dive behaviour

The top model for analysing maximum dive depth (model 1) was additive and included logger category, year, sex and time since deployment and accounted for 54% of the AICc weight (Supporting information). Compared to IgotU birds, the dive depths were slightly shallower for video loggers ( $\beta = -0.189$ ,  $p = 0.037$ ), while PathTrack birds dived

marginally deeper ( $\beta = 0.019$ ,  $p = 0.030$ ; Supporting information). Dive depths were greater in 2021 compared with 2020 ( $\beta = 0.225$ ,  $p = 0.004$ ), but there was no difference between 2022 and 2020 ( $\beta = -0.033$ ,  $p = 0.699$ , Fig. 4A). Males dived significantly deeper than females ( $\beta = 0.538$ ,  $p < 0.001$ ), and dives were slightly shallower with increasing time since logger deployment ( $\beta = -0.005$ ,  $p < 0.001$ ), but it is important to note that the effect size was very small (Supporting information).

Model 1 (logger category year, sex, and time since deployment) was also the top-supported model for both the dive duration (97% AICc weight) and bottom time (94% AICc weight) analyses (Supporting information). Birds with PathTrack loggers had significantly shorter dive times ( $\beta = -0.089$ ,  $p < 0.001$ ) and bottom times ( $\beta = -0.075$ ,  $p < 0.001$ ) compared to IgotU loggers (Fig. 4B). Year effects were similar for both metrics, with significantly longer times



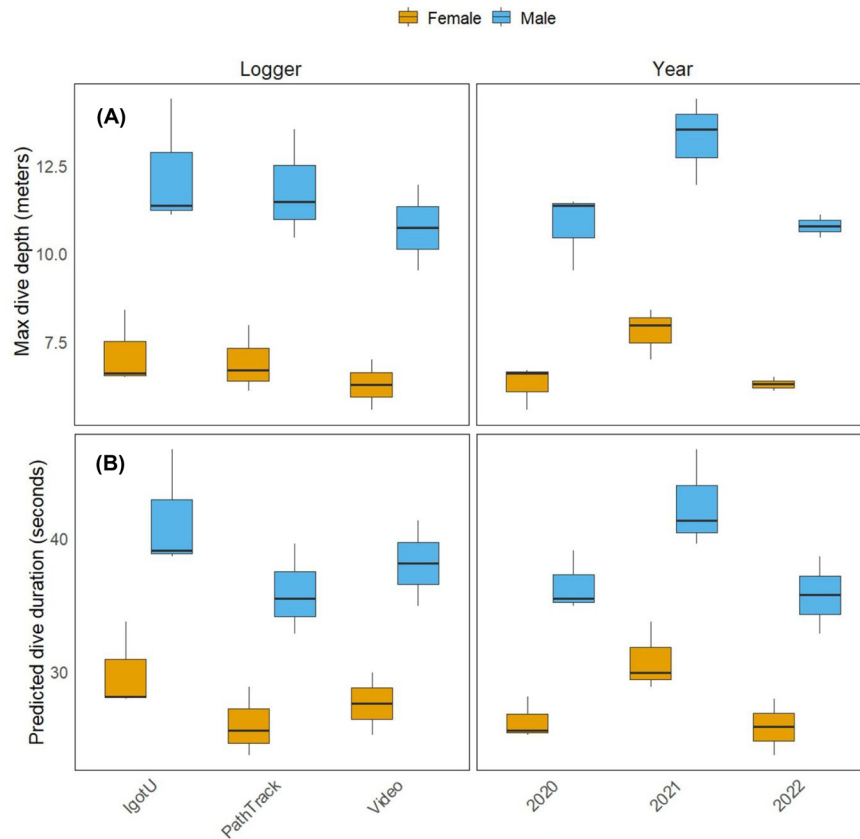


Figure 4. Predicted dive metrics from generalised linear mixed models. (A) Maximum dive depth (meters) and (B) dive duration (seconds) across three logger types and three years, shown separately for males and females. Boxplots show medians and interquartile ranges of predicted values.

in 2021 relative to 2020 (*dive time*:  $\beta=0.167$ ,  $p=0.001$ ; *bottom time*:  $\beta=0.112$ ,  $p=0.027$ ), but no difference in 2022. Males exhibited significantly longer dive and bottom times than females (*dive time*:  $\beta=0.322$ ,  $p < 0.001$ ; *bottom time*:  $\beta=0.218$ ,  $p < 0.001$ ). Finally, both dive and bottom times decreased slightly with increasing time since deployment (*dive time*:  $\beta=-0.004$ ,  $p < 0.001$ ; *bottom time*:  $\beta=-0.009$ ,  $p < 0.001$ ).

Although explanatory variables such as logger type, year, and sex had detectable effects across all of the models exploring foraging behaviour, it is important to note that a considerable amount of the variation was attributable to consistent differences among individuals (Supporting information).

### Adult body mass

Body mass at deployment (control group) showed a mean increase over the course of the breeding season (mean  $\pm$  SD: males =  $16 \pm 8$  g and females =  $18 \pm 9$  g). Contrary to this, after being fitted with IgotU, PathTrack, and video loggers, birds showed a mean loss in body mass over the period of logger deployment (Fig. 5A). Specifically, the  $\beta$  estimates demonstrated a mean reduction in body mass of 41 g for IgotU birds ( $p < 0.001$ ), 43 g for PathTrack birds ( $p < 0.001$ ), and a reduction of 55 g for video logged birds ( $p < 0.001$ ). There

was no significant difference in body mass loss between males and females in any of the logger categories ( $p=0.51$ ). It is important to consider that the duration which the data were collected over varied for each group of birds (1–4 days for IgotU loggers and video loggers, and an average of 18 days for PathTrack loggers), while the control group consisted of unpaired measurements collected in June and July from different individuals.

### Chick growth rate

Chicks from control nests increased their body mass with a median rate of  $41 \text{ g day}^{-1}$  ( $\pm 15$  SD) over the linear growth phase (day 5–30). During the period where a parent was deployed with a logger, the chicks of parents with video loggers grew at a median rate of  $8 \text{ g day}^{-1}$  ( $\pm 29$  SD), those with IgotU loggers at  $37 \text{ g day}^{-1}$  ( $\pm 25$  SD), and those with PathTrack loggers at  $45 \text{ g day}^{-1}$  ( $\pm 21$  SD; Fig. 5B). The GLM revealed significant differences in chick growth rates across logger types. Chicks of birds fitted with video loggers had substantially lower growth rates compared to the control group ( $\beta=-30.89$ ,  $p < 0.001$ ), while those with PathTrack loggers showed a slight increase in growth rate ( $\beta=5.97$ ,  $p=0.036$ ). Growth rates for chicks of birds with IgotU loggers did not differ significantly from the control group ( $\beta=-0.97$ ,  $p=0.33$ ).

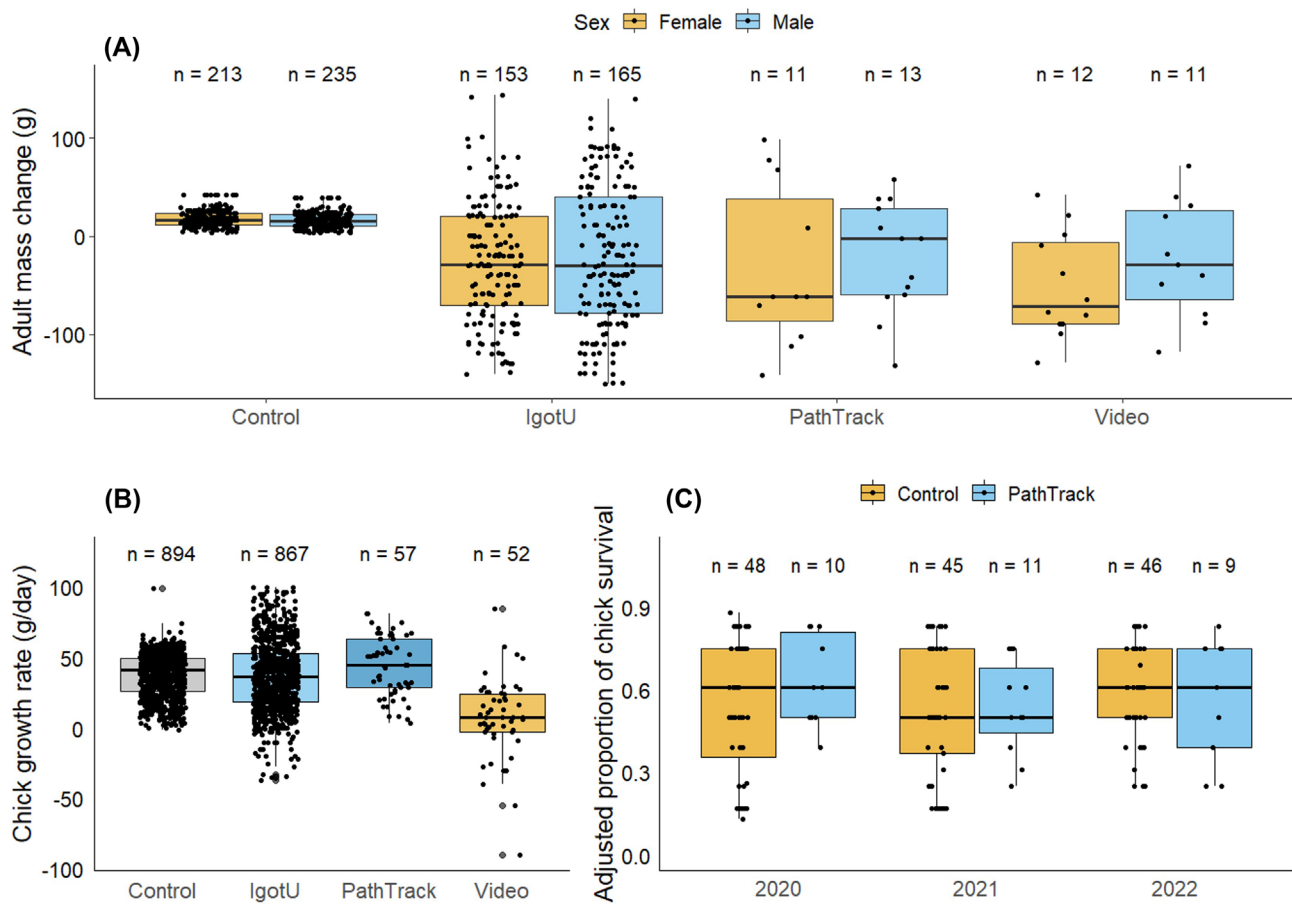


Figure 5. Effects of logger deployment on adult mass change, chick growth, and chick survival. Boxplots show medians and interquartile ranges, and points represent individual birds. (A) Mass change (g) by logger (IgotU, PathTrack, or Video) and sex. Control values are model-predicted from seasonal body mass regressions, reducing apparent variation relative to logger-equipped groups. (B) Chick growth rates are shown for control nests (no logger) versus nests with an adult fitted with a logger. (C) Adjusted proportion of chick survival is shown for long-term PathTrack deployments compared with control nests with no logger, across three breeding seasons (2020–2022).

### Reproductive success and nest failure

Reproductive success, measured as the adjusted proportion of chicks surviving per nest, did not differ between control nests (no loggers) and those with long-term PathTrack equipped birds ( $\beta = 0.01$ ,  $SE = 0.164$ ,  $p = 0.543$ ; coefficients reported on the logit scale of survival probability). Furthermore, there was no significant difference in reproductive success between 2020 (chick survival probability = 0.64) and 2022 (chick survival probability = 0.67;  $\beta = -0.08$ ,  $SE = 0.153$ ,  $p = 0.602$ ), but there was a trend toward lower reproductive success in 2021 (chick survival probability = 0.57) compared with 2022 ( $\beta = -0.240$ ,  $SE = 0.154$ ,  $p = 0.12$ ; Fig. 5C).

We did not assess reproductive success for IgotU and video logger deployments due to their short deployment duration. However, in 2020–2021, two birds with video loggers undertook > 24-hour foraging trips, leading to nest failure and preventing recapture. The male returned without its video logger several days later, while the female was not resighted (however, being only metal-ringed reduced the bird's resighting probability). In contrast, no nest failures occurred in 121 IgotU or 40 PathTrack deployments (2020–2022), though

eggs in two PathTrack nests failed to hatch for unknown reasons.

### Adult survival

Mean annual survival from year  $t$  to  $t+1$  for control birds (not fitted with loggers at year  $t$ ) was 0.83 for males (95% credible intervals (CrIs): 0.78–0.88) and 0.82 for females (95% CrIs: 0.77–0.87). For all logger categories, which included short (IgotU and short PathTrack deployments < 8 days), video and long (PathTrack deployments > 28 days), survival rates were higher than the control group (Fig. 6A). Female survival was lower than for males for individuals with video or long-term loggers (Fig. 6A). Ninety-four percent of the posterior distribution for females was lower than for males for long-term and video loggers. For short-term loggers, 48% of the posterior distribution of the mean female survival rate was lower than males. Survival rates were relatively constant over the study period although there was a tendency for a temporal decline, with estimates rising again in 2024 (Fig. 6B). Recapture rates were significantly higher for individuals captured the previous year (see 'Material and methods' on trap

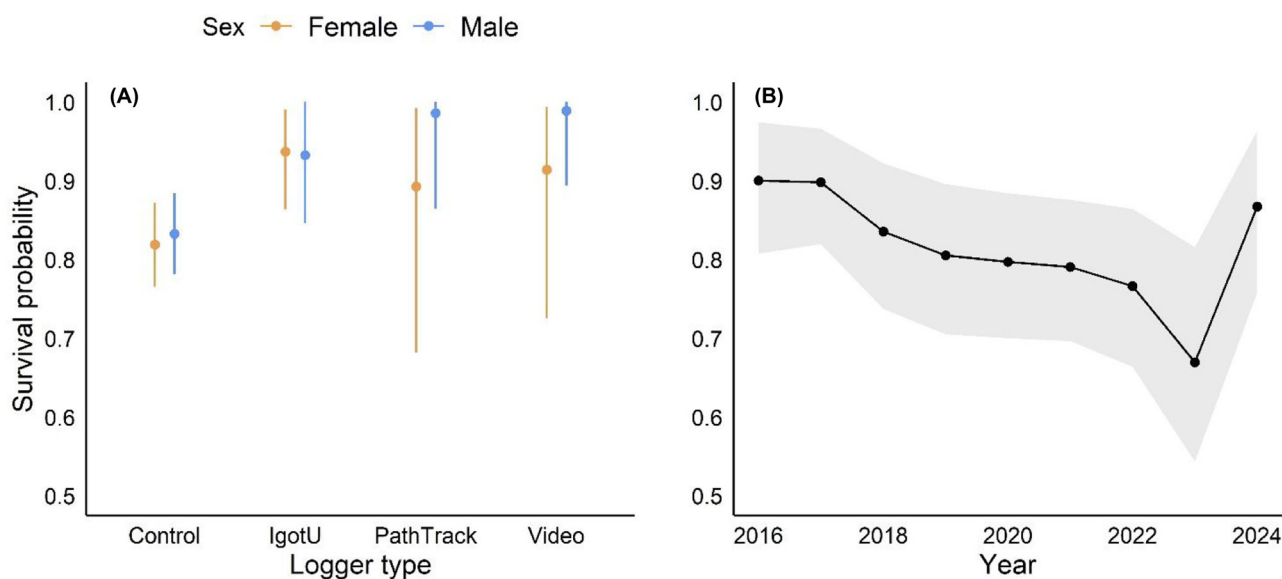


Figure 6. (A) Mean survival probabilities for individuals without a logger ('Control'), with a short-term logger ('IgotU'), a longer-term logger ('PathTrack'), or a video logger ('Video'), for females (orange) or males (blue). Vertical lines represent 95% credible intervals. (B) Annual estimates of survival rates (year t+1), for birds without a logger, where the shaded area represents 95% credible intervals.

dependence) (mean: 0.83, 95% CrIs: 0.76–0.87) than those that were not captured the previous year (mean = 0.48, 95% CrIs: 0.33–0.63).

## Discussion

Our study explored the effects of logger deployment on the behaviour, reproduction and survival of breeding shags. We observed several effects of loggers in line with our first two predictions (see summary of results Table 2): adults experienced body mass loss regardless of logger category and duration of deployment, while chick growth rates declined for parents with the heaviest loggers (video). Birds deployed with long-term PathTrack loggers and leg-mounted TDRs travelled less distance from the colony (particularly in 2021 and 2022), reduced their trip duration, and performed shorter dives compared to those with shorter deployment durations and tail-attached TDRs (IgotU), whereas birds equipped with video loggers travelled further for longer but made significantly shorter and shallower dives. Regarding our fourth prediction that females would be more affected than males, there was no evidence of deployment duration, attachment location or logger mass affecting female foraging behaviour more than males. However, there was a tendency for lower survival of females fitted with video and long-term PathTrack loggers. Consistent with our final prediction, we found a significant effect of year on foraging trip duration, dive depth, and dive duration, suggesting that birds exerted greater foraging effort in years with lower resource availability. Chick survival also declined in 2021 when the abundance of preferred prey was at its lowest during the three years (ICES 2024).

## Logger effects on foraging behaviour metrics

PathTrack birds had smaller foraging ranges and shorter dive durations than birds with IgotU loggers, despite similar logger mass and dimensions. PathTrack birds were deployed with TDRs on a leg ring for an average of 18 days and had already carried the loggers for a mean of five days when dive behaviour was analysed. In contrast, IgotU loggers were deployed for a mean of two days with the TDR attached to the tail feathers with the GPS logger. Furthermore, leg-mounted TDRs can increase drag and reduce dive efficiency, especially in foot-propelled divers like shags, which expend more energy swimming due to their propulsion method (Lovvorn and Liggins 2002, Watanuki et al. 2005). Leg rings, for example, increase energetic costs (3.5–4.3%) during diving in imperial shags *Leucocarbo atriceps*, despite the rings weighing less than 8 g (Gómez-Laich et al. 2022). Although comparatively small, these energetic costs could become considerable over the lifetime of the bird. Shorter deployments may fail to detect such effects; for instance, no difference in dive behaviour was found in great cormorants *Phalacrocorax carbo carbo* fitted with leg-mounted GLS devices for only two-to-four days (Ropert-Coudert et al. 2009). Taken together, the behavioural adjustments observed in the PathTrack birds may reflect both the cumulative effects of increased drag during diving and/or increased weight during flight, as well as spatial differences in foraging trips. In particular, the shorter model predicted distances of PathTrack birds in 2021 and 2022 likely reflect their near-colony foraging (2–4 km), in contrast to IgotU and video birds that more often travelled 18–25 km producing a bimodal distribution and variation among logger types.

The foot-propelled diving style of shags makes them more manoeuvrable at lower speeds but also necessitates greater

Table 2. Logger effects on foraging trip characteristics and fitness outcomes in seabirds, relative to IgotU (foraging) and Control (reproductive success and survival). –: no data; ≈: no difference between groups; ↓: relative decrease; ↑: relative increase; \*: year effect (2021/2022); †: sex effect present (males > females), but logger effect holds for both sexes; ‡: loggers had higher survival vs control, but females with long-term PathTrack/Video had lower survival vs males.

|                                                               | IgotU | PathTrack | Video |
|---------------------------------------------------------------|-------|-----------|-------|
| Foraging trip metrics. Baseline = IgotU                       |       |           |       |
| Max trip distance                                             | –     | ↓*        | ↑     |
| Trip duration                                                 | –     | ↓         | ↑     |
| Max dive depth                                                | –     | ↑†        | ↓†    |
| Dive duration                                                 | –     | ↓†        | ↓†    |
| Bottom time                                                   | –     | ↓†        | ↓†    |
| Reproductive success and survival metrics. Baseline = control |       |           |       |
| Adult body mass                                               | ↓     | ↓         | ↓     |
| Chick growth                                                  | ≈     | ≈         | ↓     |
| Chick survival                                                | –     | ≈         | –     |
| Adult survival                                                | ↑     | ↑‡        | ↑‡    |

foraging efficiency (Lovvorn and Liggins 2002). These are important considerations when deciding upon the duration of deployments and the attachment method. Indeed, the 3% rule focusses solely on the device weight but overlooks aerodynamic (Bowlin et al. 2010) and hydrodynamic drag (Vandenabeele et al. 2015). The impacts of these are likely to vary depending on the body size, morphology, and flight and swimming styles of the species. Although increasing logger deployment duration was associated with shallower and shorter dives, the effect sizes were relatively small, potentially scaling with time given that over 50% of our dataset comprised deployments of only 1–3 days. To investigate this further, we analysed the entire long-term PathTrack dataset, which showed that with longer deployment durations, birds travelled greater distances from the colony for longer but performed shorter dives (Supporting information). However, strong year effects, particularly from 2020, suggest that factors beyond deployment duration, such as environmental variability, may also have influenced these patterns in line with prediction four.

Birds equipped with short term video loggers (~ 4% body mass) exhibited reduced dive durations and shallower dives, consistent with findings in Manx shearwaters *Puffinus puffinus* (Clairbaux et al. 2025), yet they had the longest trip durations of all logger groups. This supports our prediction that the added mass and drag of video loggers increased energy expenditure during flight and diving, resulting in birds compensating through altered dive behaviour and extended foraging trips. Increased trip duration compared to control birds was observed in black-legged kittiwakes *Rissa tridactyla* and Manx shearwaters fitted with GPS devices > 3.7 and 4% of body mass, respectively (Heggøy et al. 2015, Gillies et al. 2020). However, our results may also reflect that the video data collection was limited to 2020–2021, with increased foraging distances in 2021 due to lower food availability. A study of southern rockhopper penguins *Eudyptes chrysocome* fitted with GPS loggers demonstrated similar responses: increased foraging trip distance and duration as well as decreased dive duration under poorer environmental conditions (Ludynia et al. 2012). However, PathTrack-equipped birds undertook significantly shorter foraging trips than those with video or

IgotU loggers, contrasting with patterns seen in black-legged kittiwakes, Manx shearwaters, and southern rockhopper penguins carrying loggers for less than seven days. This suggests that prolonged deployment may result in birds compensating for logger-induced drag and weight by reducing trip distance and duration. Our findings indicate that leg-mounted loggers, high logger mass, and extended deployment durations alter shag foraging behaviour – potentially biasing the very metrics researchers seek to measure.

### Logger effects on reproductive performance

Adult birds demonstrated a mean loss in body mass regardless of logger type. This is consistent with other seabird studies that have demonstrated a direct cost to adults (Paredes et al. 2005, Passos et al. 2010), although not observed in all studies (Heggøy et al. 2015). A laboratory study on captive pigeons fitted with loggers showed that they reduced their body mass equivalent to the weight of the devices during the deployment period, demonstrating the importance of body mass regulation in birds, even in situations when flight is restricted (Portugal and White 2022). In our study, body mass loss scaled with logger size, with the greatest mass loss of birds equipped with the heaviest video loggers and the least for equipped with the smallest loggers (IgotU). However, mass loss did not scale with attachment duration, as loss was similar between long-term PathTrack loggers and IgotU birds, where the PathTrack loggers were attached on average nine times longer than IgotU loggers. This suggests that PathTrack birds compensated for the energetic costs of logger mass and drag through means other than increased mass loss, for instance by shortening trip distances, or reduced breeding investment, potentially with support from their partners, or compromised immunocompetence (e.g. elevated corticosterone levels or H/L ratios (Heggøy et al. 2015). This could explain the lower survival of female PathTrack birds compared to IgotU birds.

Logger deployment did not impact chick growth for IgotU and PathTrack loggers or reproductive success for PathTrack loggers. However, chick growth of parents deployed with video loggers was reduced by 80% compared to the control group. These findings suggest that adults carrying PathTrack



and IgotU loggers were able to absorb the associated costs without altering their provisioning behaviour. In contrast, reduced chick growth rates, along with longer and more distant foraging trips and shallower dives, indicate that video-logged birds were unable to do so. Compensatory increases in foraging activity and chick provisioning by the partner of an instrumented bird may mitigate disturbance and energetic costs associated with logger deployment, as observed in other seabird species (Paredes et al. 2005, Heggøy et al. 2015, Symons and Diamond 2019, Gillies et al. 2020, Seward et al. 2021). In our study, the deployment duration of video loggers was limited to two days, but prolonged use at this load could pose serious risks to reproductive success and survival (Geen et al. 2019). In addition, two cases of nest failures were observed for birds with video loggers, while there were none among other logger categories or the control nests.

Finally, despite higher survival found for birds with loggers, we observed a larger discrepancy between females and males for individuals fitted with video and long-term PathTrack loggers, compared to short-term IgotU. Although survival was higher in all logger categories compared to the control group, the control group comprised individuals that did not carry a logger in a given year between 2015 and 2023. Logger birds, by contrast, were those deployed with a logger during this period, many of which were captured and colour-ringed for the first time. Birds from the logger group may thus have been on average younger, while birds from the control group were older and may thus have experienced higher mortality. Importantly, control birds were also deployed with a logger in at least one year (possibly before 2015) and were therefore bold enough to be captured twice (colour-ringed at logger retrieval), suggesting similar boldness to logger birds. If catchability/boldness is a repeatable personality trait in shags, as in other seabirds (Patrick and Weimerskirch 2014), personality likely did not bias our survival estimates. However, other aspects of individual quality or behaviour may explain the higher unexpected survival of logger birds. For IgotU and Video loggers, we selected adults that were approachable and had chicks, biasing our sample of logger birds towards early breeders, while later-breeding birds (still incubating eggs when we were deploying loggers), even colour-ringed ones, were categorised as controls. Studies on the Isle of May show selection for early laying (Keogan et al. 2021); similarly, our logger birds may have been better adapted to annual environmental conditions than later-breeding controls, potentially contributing to higher survival in the logger birds.

Nevertheless, the difference in survival probabilities between sexes remains of interest. Seabirds are typically long-lived species with high adult survival probabilities (Votier and Sherley 2017) and, thus, any reduction in breeding female survival is of great concern. Female survival was lower for those with video loggers and PathTrack loggers compared to the lighter and shorter-term IgotU loggers. This could suggest that the lighter females are more susceptible to the cumulative physiological costs of longer deployments, e.g. increased

energetic costs and/or stress responses, which can reduce body condition or reproductive success. Overall, the consequences of the heavier/longer-duration loggers were identified on the condition of adults and chicks, but both appear able to buffer any demographic consequences under these conditions, with exception of a tendency for lower female survival. Given seabirds' life history strategy, we would expect responses to first be observable in behaviour and physiology, as found here, and, if stress increases further, on reproductive success and finally adult survival (Cairns 1988, Lewis et al. 2006).

### Sex and year effects

In addition to the logger effects, we found sex related differences in foraging behaviour, with males diving to greater mean depths, and for longer, than females, likely due to males' larger body size. Our findings are consistent with two earlier studies from Sklinna that analysed dive behaviour of birds tagged between 2010–2018 with IgotU and TDR loggers (Christensen-Dalsgaard et al. 2017, Carlsen et al. 2021). This sexual segregation in foraging behaviour driven by differences in body size may allow different microhabitats to be targeted by each sex and thereby reduce competition (Harris et al. 2013, Christensen-Dalsgaard et al. 2017, Fijn et al. 2022). Greater flexibility in dive behaviour may also allow males to better buffer the energetic costs associated with logger deployment than females.

Effects also differed between study years, especially 2021, when foraging trip distance and duration, as well as dive depth and duration all increased, which coincided with lower chick survival in both logged and unlogged birds. Previous research on Sklinna demonstrated that breeding success is closely linked to the abundance of young age classes of saithe, with declines in years of low saithe abundance, when birds forage further away, dive deeper, and rely to a greater extent on less common prey (Lorentsen et al. 2019). Abundance of young saithe in 2021 and 2022 was lower than 2020 (ICES 2024), which can lead to longer foraging trips and lower reproductive success (Lorentsen et al. 2019).

### Study limitations

The study design was partly opportunistic as it used a large multi-year dataset that was not originally intended to test logger effects. For example, both partners on a nest were commonly deployed with loggers, either consecutively or with partial temporal overlap. This may have amplified impacts on chick growth in nests where both partners were tagged, introduced carry-over effects for second-deployed birds, and in some cases cross-category effects between IgotU and video loggers. PathTrack deployments were unaffected by these issues, which is important given that this was the category where we observed smaller foraging ranges and shorter dive durations.

Nests used to assess chick growth and reproductive success came from two adjacent sites. The quarry (control site) is less densely populated and has higher predation risk from white-tailed sea eagles *Haliaeetus albicilla*, potentially attracting lower quality individuals (Kokko et al. 2004), while the

breakwater (logger site) is more sheltered and densely populated, which may enhance breeding success. These site differences could independently influence outcomes, potentially underestimating any effects of logger deployment.

Finally, sex ratios in shags vary with laying date and order (Velando et al. 2002), and male and female chicks grow at different rates (Velando et al. 2000). In imperial shags, male chick growth is more affected by low food availability and the presence of younger siblings (Svage et al. 2021). An ideal design would have targeted nests with an equal number of eggs, controlled for chick sex and hatching date, and compared logger and control nests in the same area. However, we did not have the resources to genetically determine the sex of chicks in our study. Nevertheless, growth rates in control nests did not differ significantly between years (2011–2024; Supporting information), suggesting sex-ratio variation did not bias results. Moreover, logger and control nests were monitored concurrently, controlling for temporal variation in sex ratios and food availability. Thus, while sexing chicks could provide additional insights, we believe our results on logger effects on chick growth are robust.

## Conclusions and recommendations

Our results indicate that 60 g loggers (> 4% body mass) are disproportionately heavy for shags and may substantially increase drag. Moreover, birds equipped with longer-term, leg-mounted TDRs showed reduced trip distances and shorter dives compared to those carrying tail-mounted GPS–TDRs of the same weight for only two days. Given recent technological advances, we recommend using GPS loggers with integrated depth sensors to eliminate the need for leg-mounted TDRs. However, leg-mounted loggers are still widely used to track shags in the non-breeding season and our findings highlight the need for careful assessment of their effects. Our results also indicate that longer deployment durations may impact female survival. Based on this, we recommend minimising the number of birds used in long-term tracking studies and avoiding repeated logger deployments on the same individuals in consecutive years. Regardless of logger type, all deployed devices resulted in mass loss in adults, while the heaviest video loggers significantly reduced chick growth rates. Our findings in shags confirm that even small devices (e.g. < 3% body mass) can have measurable effects, highlighting the need for control studies and systematic reporting of device effects across species, attachment methods, and deployment durations (Barron et al. 2010, Bodey et al. 2018, Cleasby et al. 2021). Researchers should also consider that, in addition to potential device effects, handling animals causes stress and hence studies should be designed to minimise impacts, while maximising output (e.g. by sharing data). Biologging has greatly advanced our understanding of seabird ecology, informing conservation and resource management. By refining our methodology and standardising the reporting of logger effects, we can continue

to gather valuable ecological data while minimising adverse impacts on the study species.

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**Conflict of interest** – The authors declare no conflict of interest.

**Permits** – We obtained permits to work on Sklinna by the local county governor, permissions to catch birds and attach loggers from the Environmental Agency and The Norwegian Food Safety Authority (Mattilsynet). The animal study was reviewed and approved by The Norwegian Food Safety Authority (Mattilsynet).

## Author contributions

**Graeme Shannon:** Conceptualization (equal); Formal analysis (lead); Investigation (equal); Methodology (equal); Visualization (lead); Writing – original draft (lead); Writing – review and editing (lead). **Kate Layton-Matthews:** Formal analysis (supporting); Visualization (supporting); Writing – original draft (supporting); Writing – review and editing (supporting). **Svein-Håkon Lorentsen:** Data curation (supporting); Investigation (supporting); Methodology (supporting); Project administration (supporting); Writing – review and editing (supporting). **Jenny Mattisson:** Formal analysis (supporting); Methodology (supporting); Writing – review and editing (supporting). **Signe Christensen-Dalsgaard:** Conceptualization (equal); Formal analysis (supporting); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Writing – original draft (supporting); Writing – review and editing (supporting). **Nina Dehnhard:** Conceptualization (equal); Data curation (lead); Formal analysis (equal); Funding acquisition (lead); Investigation (lead); Methodology (equal); Project administration (lead); Writing – original draft (equal); Writing – review and editing (equal).

## Transparent peer review

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/jav.03534>.

## Data availability statement

Data are available from the Dryad Digital Repository: <https://doi.org/10.5061/dryad.2z34tmq07> (Shannon et al. 2025).

## Supporting information

The Supporting information associated with this article is available with the online version.

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